

# Harshana Lakshara Fernando

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## SUMMARY

Software Engineer specializing in Distributed Systems, Microservices, and Event-Driven Architecture, with hands-on experience building Kafka-based backend services and optimizing production system performance. Proficient in Java, Spring Boot, Go, gRPC, REST APIs, PostgreSQL, Docker, Kubernetes, CI/CD, and observability using Prometheus and Grafana.

## EXPERIENCE

### Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland

Remote

- Built Go/Gin-based backend services for wearable time-series ingestion and cross-device synchronization.
- Designed PostgreSQL schemas for participants, studies, devices, consent, and sensor streams.
- Integrated Keycloak OAuth2/OIDC with Traefik as API gateway and load balancer, via Docker Compose.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an HL7/FHIR-aligned schema mapping participants, devices, and wearable measurements.

### Software Engineer Intern

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd — Aviation Technology

Colombo, Sri Lanka

- Built a distributed Spring Boot/Kafka-backed PNR ID generator, live in production, achieving 25+ PNRs/sec at <100ms latency.
- Resolved Kafka consumer lag and offset issues using shutdown hooks, retry handling, and a Producer-Consumer refactor.
- Integrated gRPC clients, Kafka consumers, and feature toggles into a legacy Java 8 AeroMart monolith.
- Implemented database-driven feature toggles for granular PNR routing during a phased production rollout.
- Maintained >90% unit test coverage while delivering production-ready microservice features.
- Deployed Prometheus and Grafana dashboards for queue-time, latency, and service monitoring.

### Freelance Software Engineer

May 2025 – May 2026

Client Platforms

Remote / Sri Lanka

- Built Agrinova, a multi-branch inventory and sales platform automating legacy manual inventory workflows.
- Implemented transaction-safe stock logic across 3 workflows: material issuance, returns, and purchase operations.
- Optimized PostgreSQL queries/indexing and improved dashboard responsiveness using Next.js caching and Zod validation.
- Developed TipLanka, a low-latency creator tipping platform with live PayHere webhook integration.
- Architected cost-efficient serverless infrastructure using Vercel and Supabase/PostgreSQL.

## PROJECTS

### Employee Management System [Architecture]

Java 17, Spring Boot, Spring Cloud, Kubernetes

- Built cloud-native Spring Boot microservices with Eureka, Config Server, API Gateway, and RabbitMQ-based config refresh.
- Implemented Circuit Breaker, Retry, and Rate Limiter patterns with Resilience4j, plus CI/CD and monitoring.

### ZK-Rollup L2 Benchmarking Framework

Rust, Solidity, TypeScript, Python, Docker

- Built an end-to-end benchmark pipeline with Rust Submitter, Solidity L1 contracts, and Python metrics tools.
- Raised Submitter test coverage from 41% to 88% and measured ~70% lower DA cost with blob-style mode.

### Maison Vella — E-Commerce Admin Panel [Architecture]

Next.js 14, tRPC, Prisma, PostgreSQL, Sanity

- Built a responsive admin dashboard using Next.js 14, Server Actions, ShadCN UI, and Tailwind CSS.
- Implemented type-safe data access with tRPC, Prisma, Supabase/PostgreSQL, Sanity CMS, and Clerk Auth.

### Serverless Auction Platform [Architecture]

Node.js, AWS Lambda, API Gateway, DynamoDB, SQS/SNS

- Architected 3 event-driven services for auctions, auth, and notifications using AWS Lambda and EventBridge.
- Designed DynamoDB GSIs and secured APIs with a custom JWT Lambda Authorizer and Middy validation.

### GoTogether Travel Platform [Architecture]

Next.js, Go, Spring Boot, Keycloak, PostgreSQL

- Built a full-stack travel platform with Next.js, Spring Boot social services, and Go/gRPC itinerary APIs.
- Configured OAuth2/OIDC with Keycloak and deployed backend services on Oracle Cloud with NGINX/Certbot.

### Food Delivery Application

Java, Spring Boot, gRPC, Kafka, Redis, WebSockets

- Developed the Restaurant service with JWT authentication, CRUD APIs, and Redis caching with invalidation.
- Enabled real-time order sync using Kafka pub/sub and WebSockets for restaurant/rider notifications.

## SKILLS

Backend: Java, Spring Boot, Spring Cloud, Microservices, Node.js, Go, Gin, REST, gRPC, GraphQL

Frontend: React, Next.js, JavaScript, TypeScript

Messaging: Kafka, RabbitMQ, WebSockets, Protobuf

Data: PostgreSQL, MySQL, Oracle DB, Redis, Couchbase

Cloud / DevOps: AWS, Docker, Kubernetes, GitHub Actions, Jenkins

Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest

## EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa

2022 – 2026

G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

## CERTIFICATIONS

CKA Training (KodeKloud, 2025) · AWS SAA-C03 Training (Udemy, 2025) · Apache Kafka Series (I, II, III) · 100 Days of DevOps (KodeKloud, 2025) · Advanced OAuth Security (Udemy, 2024)